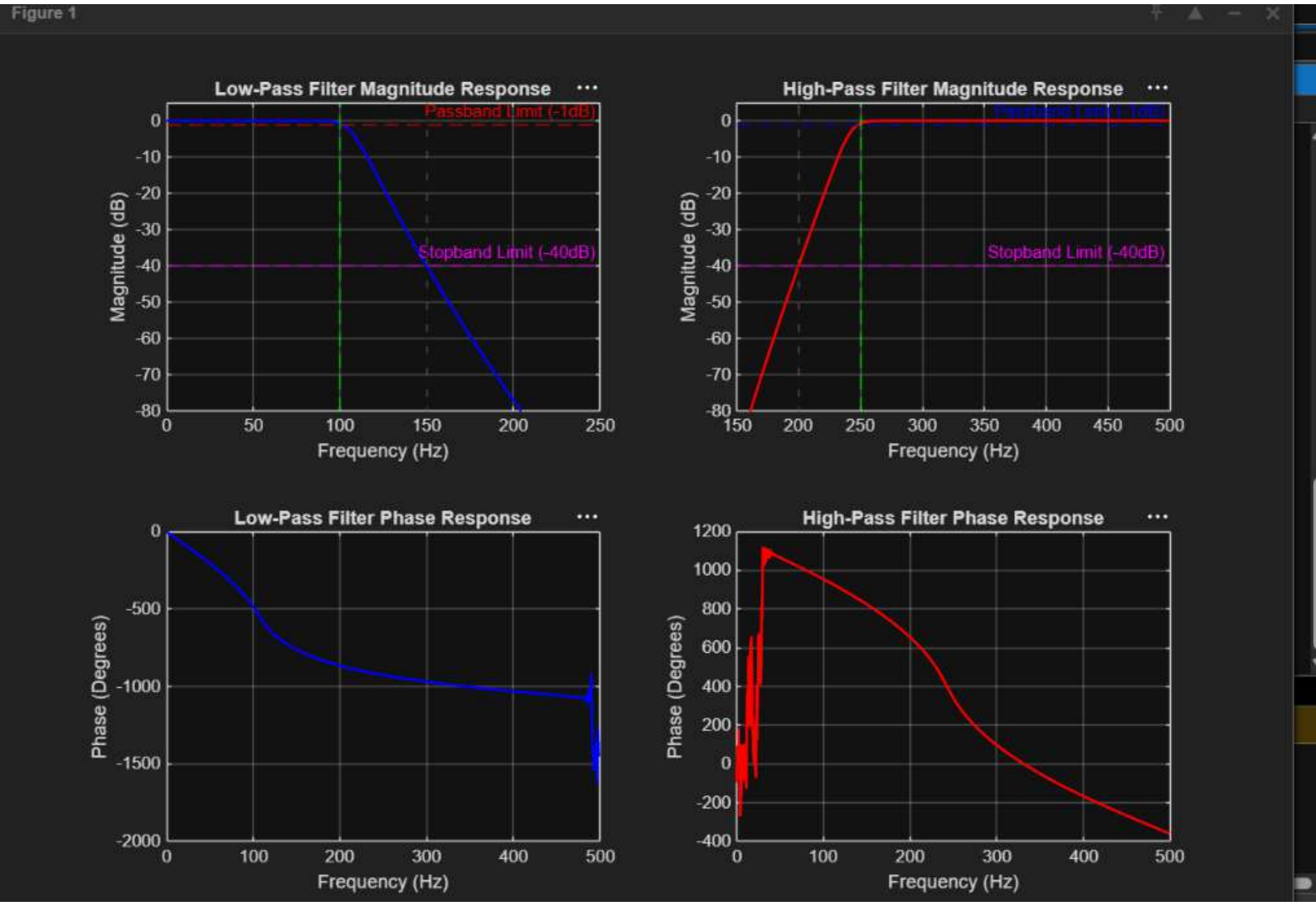
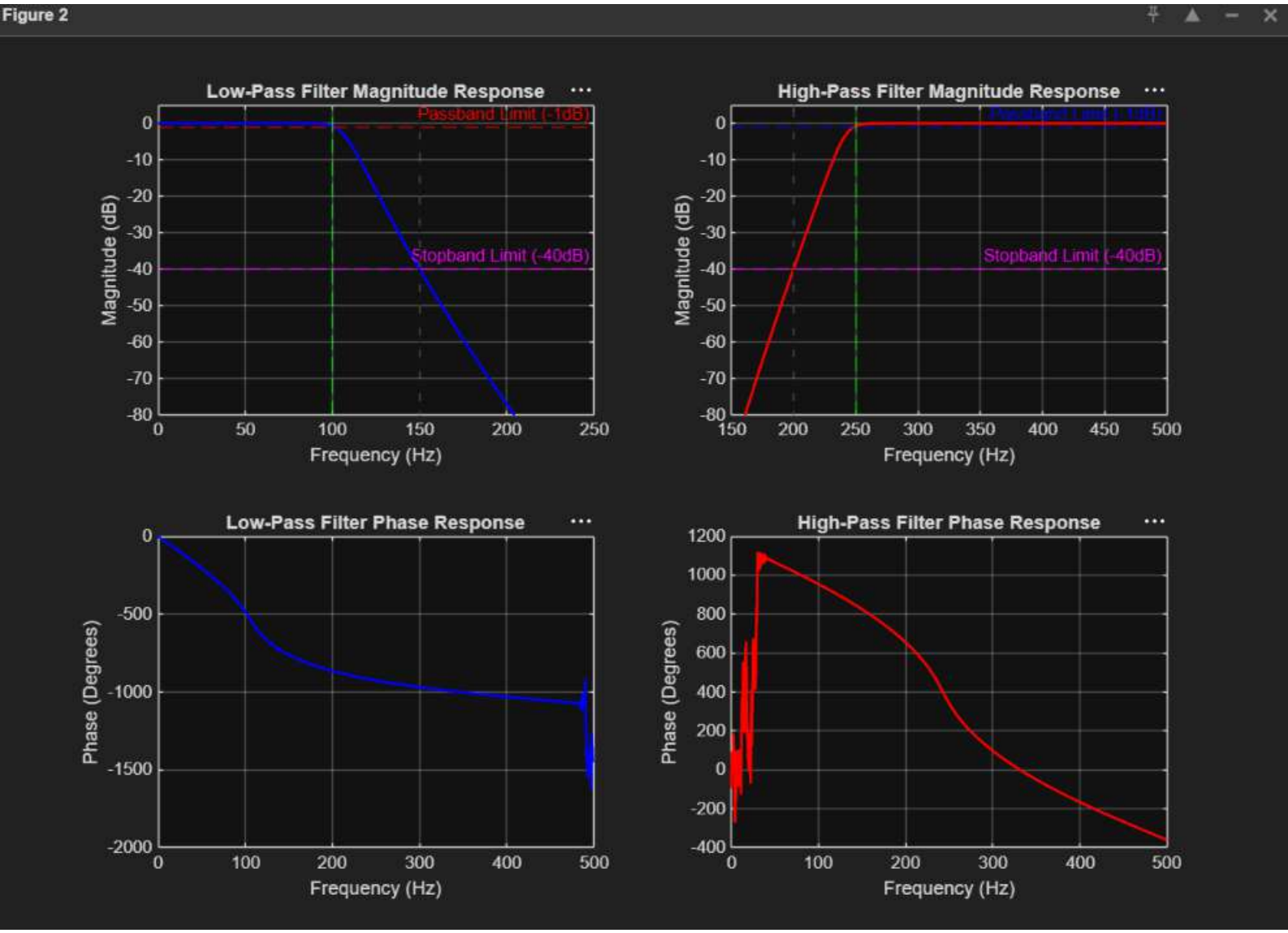
```
Fs = 1000;
Fp_LP = 100;
Fst_LP = 150;
Fst_HP = 200;
Fp_HP = 250;
Ap = 1;
Ast = 40;
Wp_LP = Fp_LP / (Fs/2);
Wst_LP = Fst_LP / (Fs/2);
Wst_HP = Fst_HP / (Fs/2);
Wp_HP = Fp_HP / (Fs/2);
[N_LP, Wc_LP] = buttord(Wp_LP, Wst_LP, Ap, Ast);
[b_LP, a_LP] = butter(N_LP, Wc_LP, 'low');
[H_LP, f] = freqz(b_LP, a_LP, 2048, Fs);
[N_HP, Wc_HP] = buttord(Wp_HP, Wst_HP, Ap, Ast);
[b_HP, a_HP] = butter(N_HP, Wc_HP, 'high');
[H_HP, ~] = freqz(b_HP, a_HP, 2048, Fs);
fprintf('-------------------------------------------------\n');
fprintf('           BUTTERWORTH IIR FILTER DESIGN         \n');
fprintf('-------------------------------------------------\n');
fprintf('Low-Pass Filter Order (N_LP)    : %d\n', N_LP);
fprintf('Low-Pass Cutoff Freq (Fc_LP)    : %.2f Hz\n', Wc_LP * (Fs/2));
fprintf('High-Pass Filter Order (N_HP)   : %d\n', N_HP);
fprintf('High-Pass Cutoff Freq (Fc_HP)   : %.2f Hz\n\n', Wc_HP * (Fs/2));
disp('LPF Numerator Coefficients (b_LP):'); disp(b_LP);
disp('LPF Denominator Coefficients (a_LP):'); disp(a_LP);
disp('HPF Numerator Coefficients (b_HP):'); disp(b_HP);
```

```
disp('HPF Denominator Coefficients (a_HP):'); disp(a_HP);
figure('Position', [100, 100, 1000, 700]);
subplot(2,2,1);
plot(f, 20*log10(abs(H_LP)), 'b', 'LineWidth', 1.5); hold on;
grid on; yline(-1, 'r--', 'Passband Limit (-1dB)'); yline(-40, 'm--', 'Stopband Limit (-40dB)');
xline(Fp_LP, 'g--'); xline(Fst_LP, 'k--');
title('Low-Pass Filter Magnitude Response');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
ylim([-80 5]);
subplot(2,2,3);
plot(f, unwrap(angle(H_LP))*180/pi, 'b', 'LineWidth', 1.5);
grid on;
title('Low-Pass Filter Phase Response');
xlabel('Frequency (Hz)'); ylabel('Phase (Degrees)');
subplot(2,2,2);
plot(f, 20*log10(abs(H_HP)), 'r', 'LineWidth', 1.5); hold on;
grid on; yline(-1, 'b--', 'Passband Limit (-1dB)'); yline(-40, 'm--', 'Stopband Limit (-40dB)');
xline(Fst_HP, 'k--'); xline(Fp_HP, 'g--');
title('High-Pass Filter Magnitude Response');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
ylim([-80 5]);
subplot(2,2,4);
plot(f, unwrap(angle(H_HP))*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('High-Pass Filter Phase Response');
xlabel('Frequency (Hz)'); ylabel('Phase (Degrees)');
```


Figure 1
Low-Pass Filter Magnitude Response
Passband Limit (-1dB)
Stopband Limit (-40dB)
Magnitude (dB)
Frequency (Hz)
High-Pass Filter Magnitude Response
Passband Limit (-1dB)
Stopband Limit (-40dB)
Magnitude (dB)
Frequency (Hz)
Low-Pass Filter Phase Response
Phase (Degrees)
Frequency (Hz)
High-Pass Filter Phase Response
Phase (Degrees)
Frequency (Hz)


Figure 2
Low-Pass Filter Magnitude Response
Passband Limit (-1dB)
Stopband Limit (-40dB)
Magnitude (dB)
Frequency (Hz)
High-Pass Filter Magnitude Response
Passband Limit (-1dB)
Stopband Limit (-40dB)
Magnitude (dB)
Frequency (Hz)
Low-Pass Filter Phase Response
Phase (Degrees)
Frequency (Hz)
High-Pass Filter Phase Response
Phase (Degrees)
Frequency (Hz)